SafeMine v2 — Landmine Detection Research Summary

Generated 2026-10-15 · AeroGuard · frozen v2 split · full provenance in RUN_LOG.md / DECISIONS.md

Headline results

System	Key metric	Value
YOLOv8s (3 seeds)	test mAP50 (mean ± std)	0.9768 ± 0.0034
Two-stage arm C (seed 0)	FP/img @ R≈0.96	0.153
Two-stage arm C (seed 0)	precision	0.894
Two-stage C−B bootstrap	Δprecision CI95	[0.065, 0.186] (excl. 0)
Grad-CAM saliency	energy enrichment (YOLO)	86× (vs MNv3 1.8×)
Label audit	objective test-box errors fixed	2 (ds0_0801, ds0_0883)

T1. Dataset & leakage audit

· split · images · instances · negatives · sources ·
· --- · --- · --- · --- · --- · --- ·
· train · 3907 · 11550 · 20 · Jan_lwir:100, May_lwir:776, ds0:2589, ds1:442 ·
· valid · 181 · 419 · 7 · Jan_lwir:42, ds0:138, ds1:1 ·
· test · 189 · 253 · 5 · Jan_lwir:10, ds0:151, ds1:28 ·
Exact cross-split duplicates: 0.
_Near-dup candidates (pHash ham<=2 vs train): 99 — all residual MSE>=958.6 after the second-method filter (0 true leaks; 30 leaked eval images removed pre-freeze; D014/D015). Manifest sha256 in split_v2/sha256.txt._
_Base-key collisions across splits: {'train_valid': 0, 'train_test': 0, 'valid_test': 0}._

T2. YOLO operating point (3 seeds)

· model · params · GFLOPs · weights MB · latency ms (med) · val mAP50 · test mAP50 · test R@OP · test P@OP · FP/img@OP · OP conf (val) ·
· --- · --- · --- · --- · --- · --- · --- · --- · --- · --- · --- · --- ·
· yolov8s · 11,125,971 · 28.6 · 21.5 · 6.5 · 0.9544 ± 0.0027 · 0.9768 ± 0.0034 · 0.9763 ± 0.0040 · 0.5157 ± 0.2180 · 1.510 ± 1.009 · 0.02/0.01/0.21 ·
_n/m runs out of scope under the approved reduced budget (D-plan scope); only yolov8s was trained on v2 (3 seeds). Metrics are mean ± std over seeds 0/1/2. Latency: PC GPU FP32, imgsz 640 (size_latency.json, seed 0). OP = recall-first conf chosen per seed on val (conf listed per seed)._

T3. Image-level comparison

· system (test, image level) · accuracy · balanced acc · precision · recall · kappa ·
· --- · --- · --- · --- · --- · --- · --- ·
· MNv3 A (thr 0.5) · 0.974 · 0.500 · 0.974 · 1.000 · 0.000 ·
· MNv3 A (recall-first thr) · 0.968 · 0.497 · 0.973 · 0.995 · -0.009 ·
· MNv3 A+B (thr 0.5) · 0.974 ± 0.000 · 0.500 ± 0.000 · 0.974 ± 0.000 · 1.000 ± 0.000 · 0.000 ± 0.000 ·
· MNv3 A+B (recall-first thr) · 0.968 ± 0.005 · 0.562 ± 0.056 · 0.977 ± 0.003 · 0.991 ± 0.006 · 0.156 ± 0.148 ·
· YOLO deploy (conf from val) · 0.984 ± 0.009 · 0.830 ± 0.115 · 0.991 ± 0.006 · 0.993 ± 0.008 · 0.677 ± 0.199 ·
· YOLO op95 (conf from val) · 0.986 ± 0.003 · 0.993 ± 0.002 · 1.000 ± 0.000 · 0.986 ± 0.003 · 0.784 ± 0.038 ·
· Two-stage (arm C, 3 seeds) · 0.991 ± 0.003 · 0.995 ± 0.002 · 1.000 ± 0.000 · 0.991 ± 0.003 · 0.854 ± 0.045 ·
MNv3 A+B = mean ± std over seeds 0/1/2; MNv3 A is seed 0; YOLO deploy/op95 = mean ± std over seeds 0/1/2 (each seed's OP conf chosen on val).
MNv3 recall-first thresholds chosen on val only.

T4. Two-stage ablation

· arm · conf/thr · precision · recall · F2 · FP/img · img recall ·
· --- · --- · --- · --- · --- · --- · --- ·
· A (YOLO deploy) · 0.02 · 0.427 ± 0.000 · 0.976 ± 0.000 · 0.777 ± 0.000 · 1.751 ± 0.000 · 1.000 ± 0.000 ·
· B (YOLO matched recall) · 0.125 · 0.762 ± 0.000 · 0.972 ± 0.000 · 0.921 ± 0.000 · 0.407 ± 0.000 · 0.995 ± 0.000 ·
· C (two-stage) · yolo 0.05, P=0.4, t=0.27 · 0.879 ± 0.018 · 0.964 ± 0.000 · 0.946 ± 0.004 · 0.178 ± 0.030 · 0.991 ± 0.003 ·
Paired bootstrap (seed 0): C-B precision +0.128 CI [0.065, 0.186]; FP/img -0.253 CI [-0.460, -0.095]; recall -0.0079 CI [-0.0195, 0.0000].
Cost (plan §5.4, seed 0): 16.0 ms/image end-to-end, 3.06 crops/image (p95 3.6), PC GPU FP32.
Means ± std over seeds 0/1/2 for arm C; A/B are seed-independent except threshold choice (B picked per seed on val).

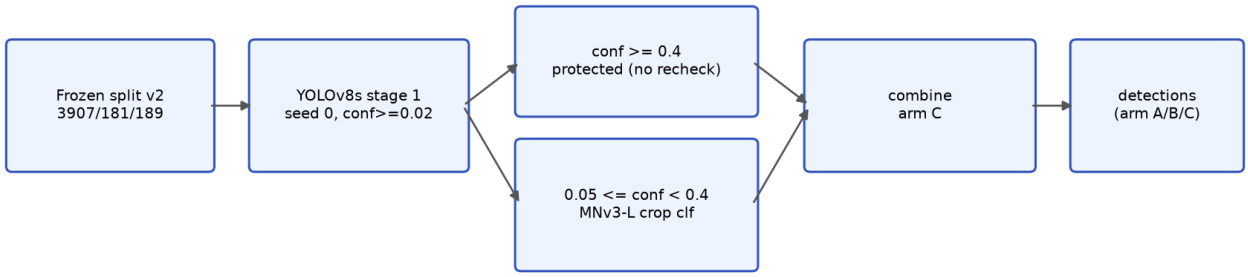
T5. Grad-CAM experiments

· experiment · metric · value ·
· --- · --- · --- ·
· E1 energy-in-box (n=244) · median energy share · 0.939 ·
· E1 · mean energy share · 0.690 ·
· E1 · median enrichment · 58.2x ·
· E1 · peak inside rate · 0.730 ·
· E2 deletion (n=182) · mean drop, mine masked · 0.739 ·
· E2 · mean drop, background masked · 0.0015 ·
· E6 head-to-head · YOLO enrichment median · 86.0x ·
· E6 · MNv3 baseline enrichment median · 1.82x ·
· E6 · paired diff (Y-M) 95% CI · [125.4, 170.0] ·
· E6 · stage-2 crop enrichment median · 2.13x ·
Grad-CAM shows which regions influenced the prediction; it does not prove the prediction is correct (plan wording rule).

T6. Failure analysis

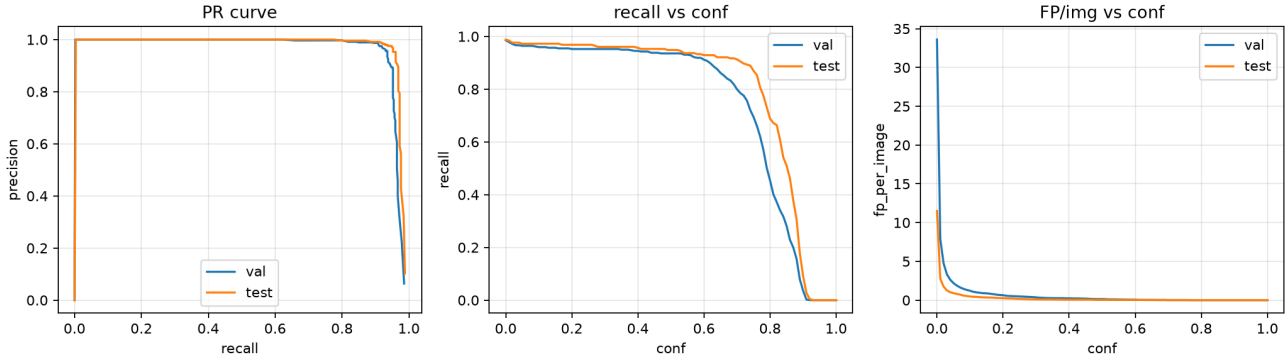
· category · FN · FP ·
· --- · --- · --- ·
· camouflaged_or_low_contrast · 5 · 0 ·
· fence · 0 · 1 ·
· label_error · 2 · 0 ·
· low_light_ground · 0 · 2 ·
· sensor_noise · 0 · 1 ·
· small · 2 · 0 ·
· structure_edge · 0 · 1 ·
· structure_wall · 0 · 1 ·
· terrain_texture · 0 · 22 ·
· tree_trunk · 0 · 1 ·
· **total** · **9** · **29** ·
_FN at the deployment arm-C threshold; FP capped at 50 (29 present). Sources: e4_failures.csv + e4_fp_sheet.png._

Two-stage pipeline

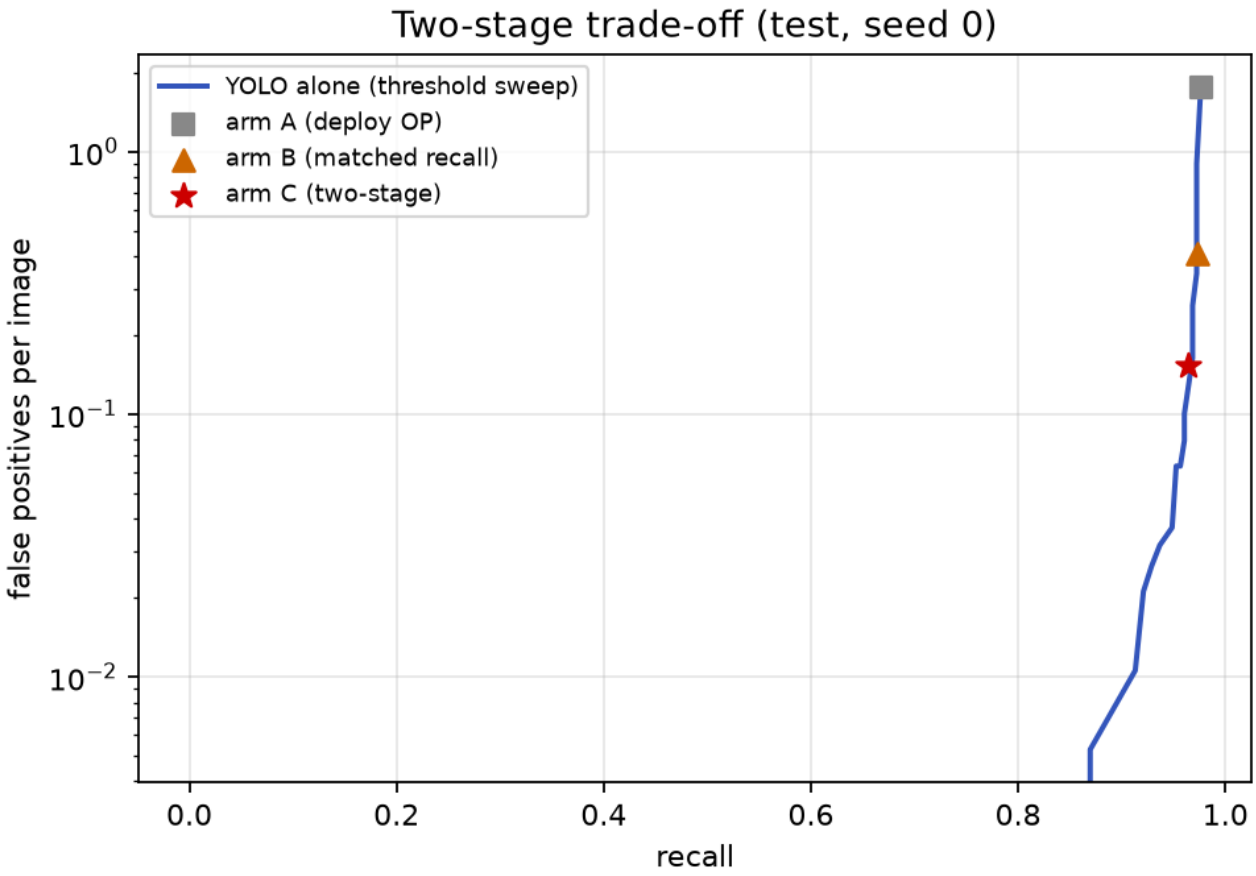


crop = box + 20% context, 224x224, sigmoid score; tau and P chosen on val only

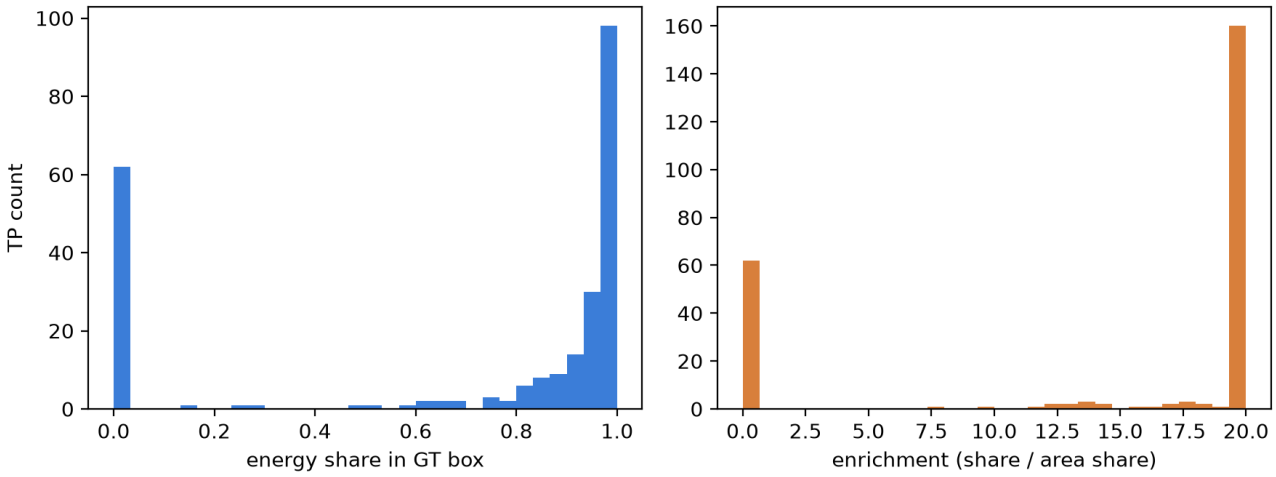
PR / recall / FP curves (seed 0)



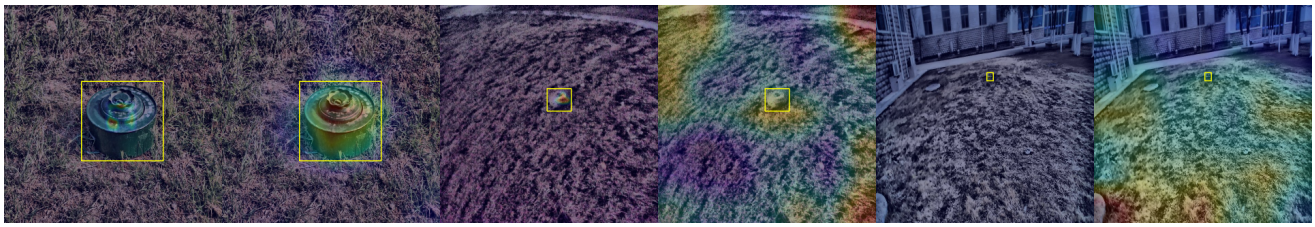
FP/img vs recall (plan §5.4)



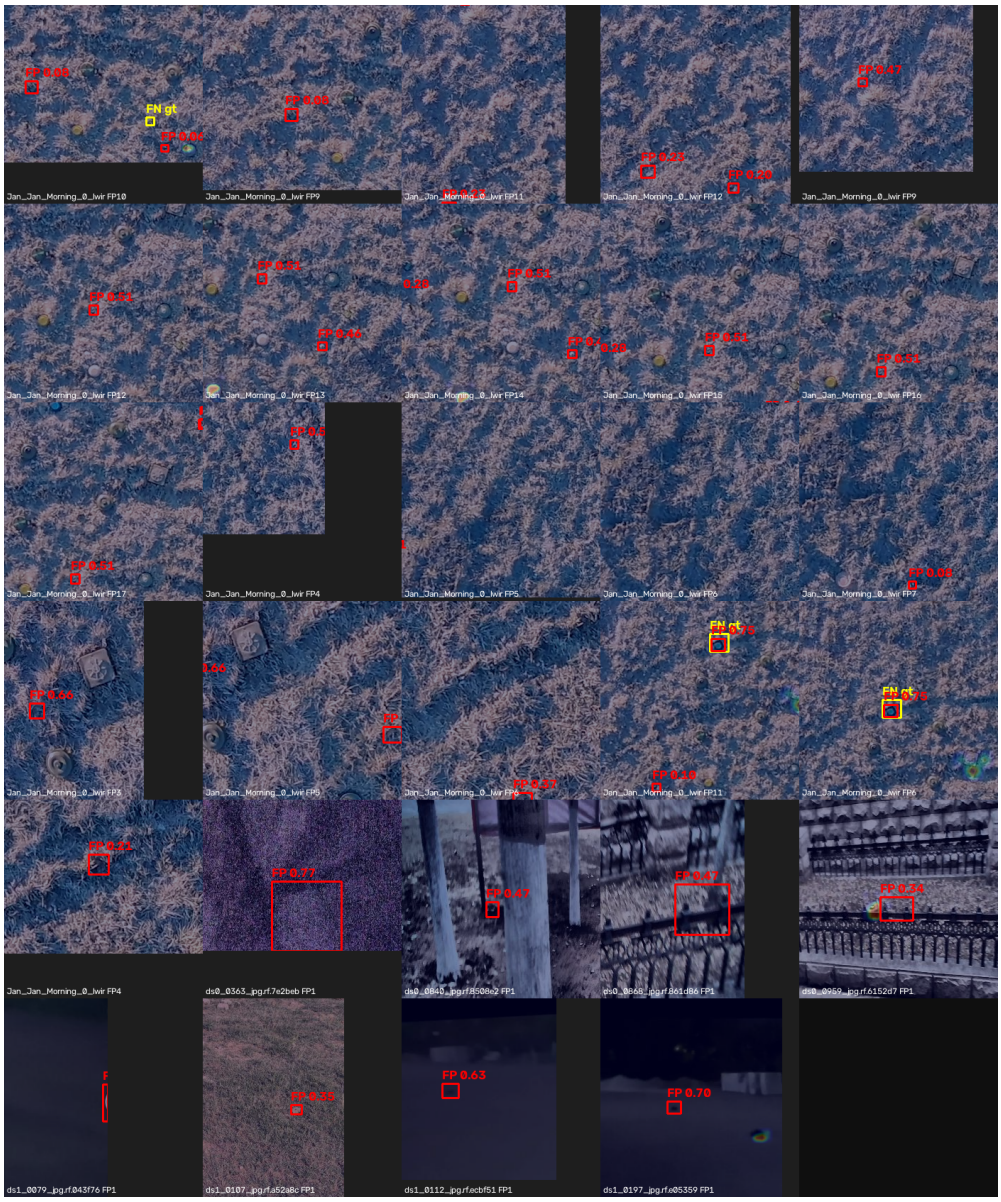
Grad-CAM energy in GT boxes (E1)



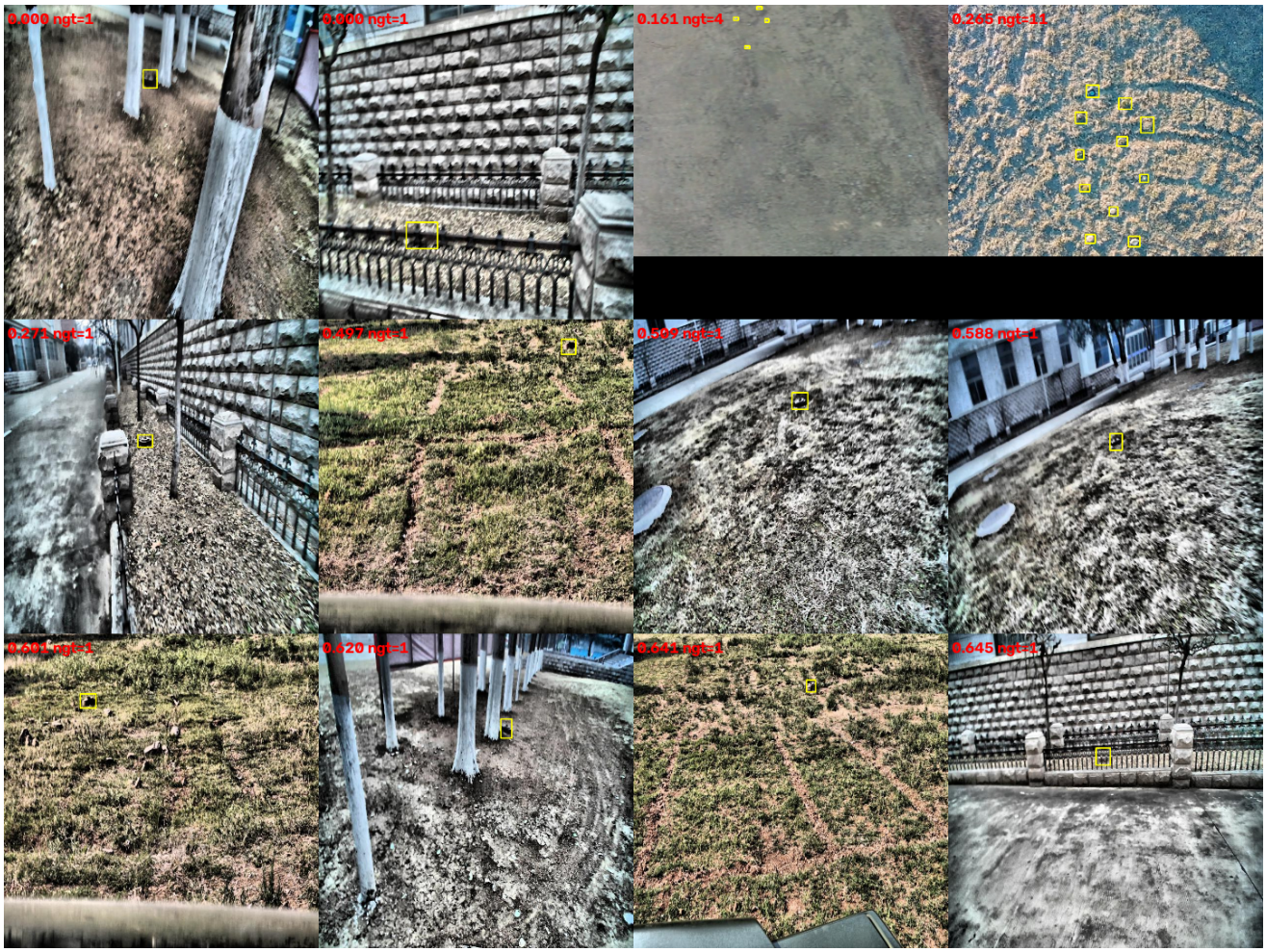
Grad-CAM head-to-head (E6)



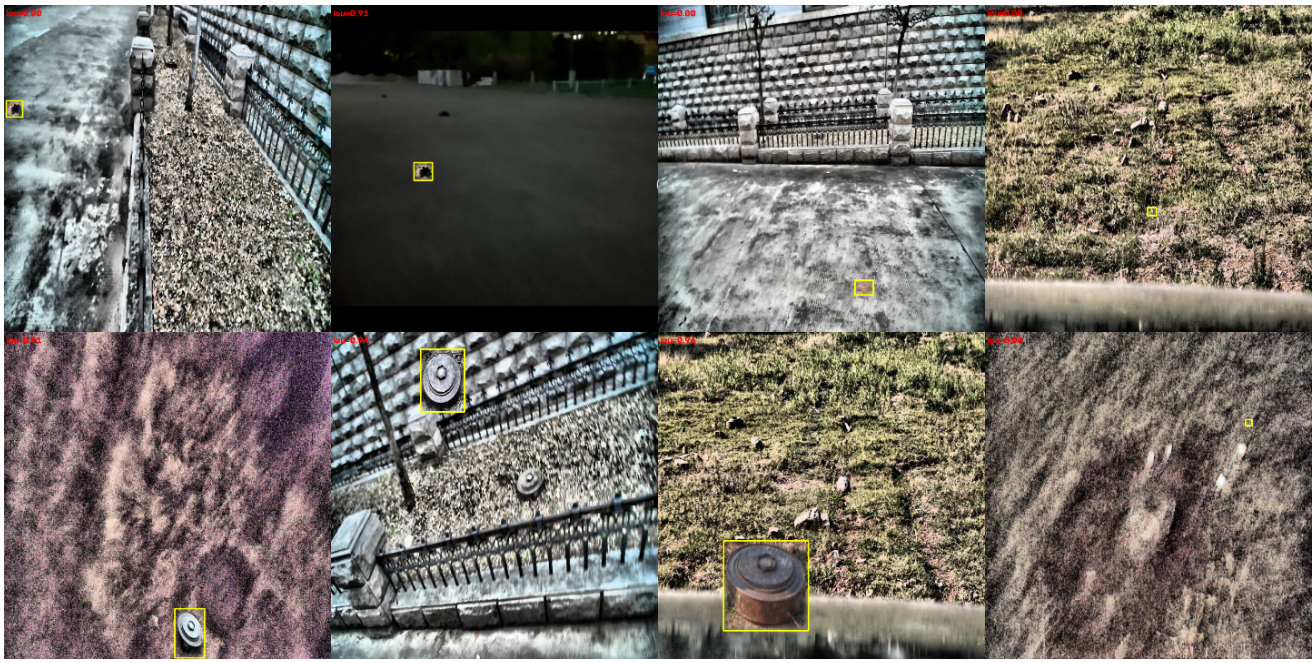
FP failure gallery (E4)



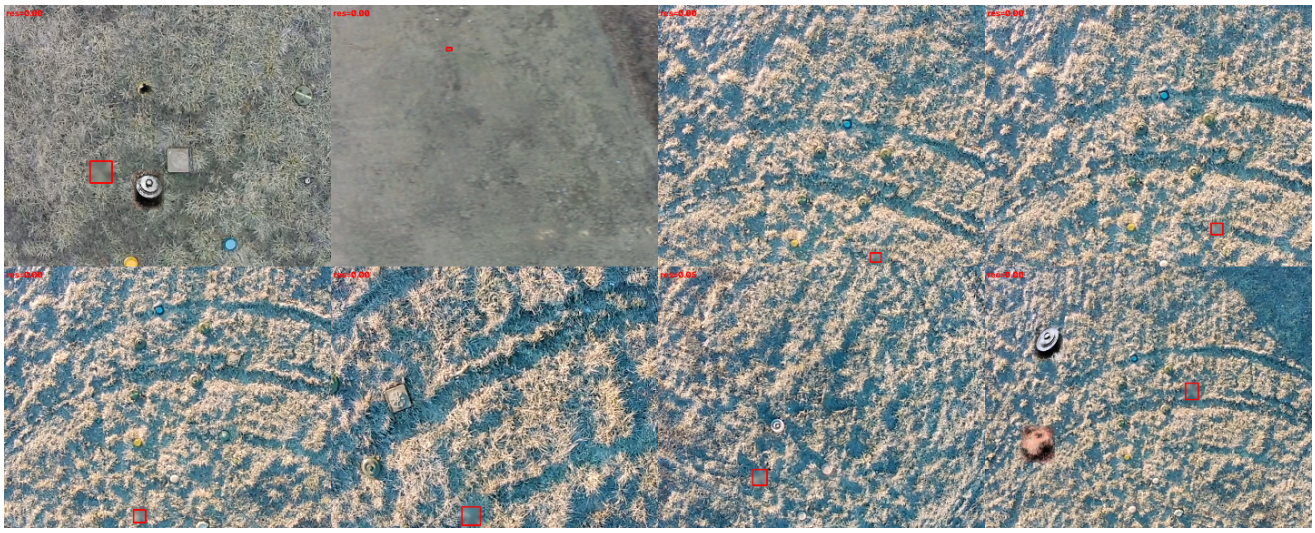
E5 zero-energy audit



E3 paste-hold counterfactual



E3 remove-disappear counterfactual



Per-source false positives (seed 0)

Per-source test results (source = dataset origin, from split_manifest.csv)
A_yolo_only_deploy
· source · images · TP · FP · FN · precision · recall · FP/img ·
· --- · --- · --- · --- · --- · --- · --- · --- ·
· Jan_lwir · 10 · 73 · 301 · 4 · 0.195 · 0.948 · 30.100 ·
· ds0 · 151 · 146 · 21 · 2 · 0.874 · 0.986 · 0.139 ·
· ds1 · 28 · 28 · 9 · 0 · 0.757 · 1.000 · 0.321 ·
· ALL · 189 · 247 · 331 · 6 · 0.427 · 0.976 · 1.751 ·
C_two_stage
· source · images · TP · FP · FN · precision · recall · FP/img ·
· --- · --- · --- · --- · --- · --- · --- · --- ·
· Jan_lwir · 10 · 70 · 21 · 7 · 0.769 · 0.909 · 2.100 ·
· ds0 · 151 · 146 · 4 · 2 · 0.973 · 0.986 · 0.026 ·
· ds1 · 28 · 28 · 4 · 0 · 0.875 · 1.000 · 0.143 ·
· ALL · 189 · 244 · 29 · 9 · 0.894 · 0.964 · 0.153 ·

Label fixes (D023)

Two objectively wrong test boxes corrected (ds0_0801_jpg.rf.7fec8a048a84003b9dcad3f5ee1eb201.jpg, ds0_0883_jpg.rf.5a0b82d51317a21637b5f97493e77bcf.jpg). Arm C recall 0.9644 → 0.9721; image recall 0.9891 → 1.0000; FP/img unchanged at 0.153. See paper_results/label_fix_report.json for full numbers.

Scope notes

Approved reduced budget: yolov8s ×3 seeds only (no n/m size rows); PC GPU only (Jetson/T7 edge phases dropped); no retrain on fixed train labels (none found). Dataset not in git — v2 split manifest sha256 in paper_results/split_v2/sha256.txt. Weights in models/ (see models/README.md).